

INTRODUCTION

Finding parking at ATU is a common issue for students, especially during busy periods of the academic year when spaces are limited. This can lead to wasted time and added stress around campus car parks.

This project explores how algorithms and computer vision can be used in a smart parking system to detect available and occupied spaces, then guide drivers to the nearest free spot. It provides a practical and low cost alternative to traditional sensor based parking solutions.



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THE SOLUTION

The proposed solution uses an overhead camera and a YOLOv8 model trained on aerial footage to analyse a car park image and determine which spaces are occupied. Predefined parking spot polygons are used to compare detections against known parking locations in real time.

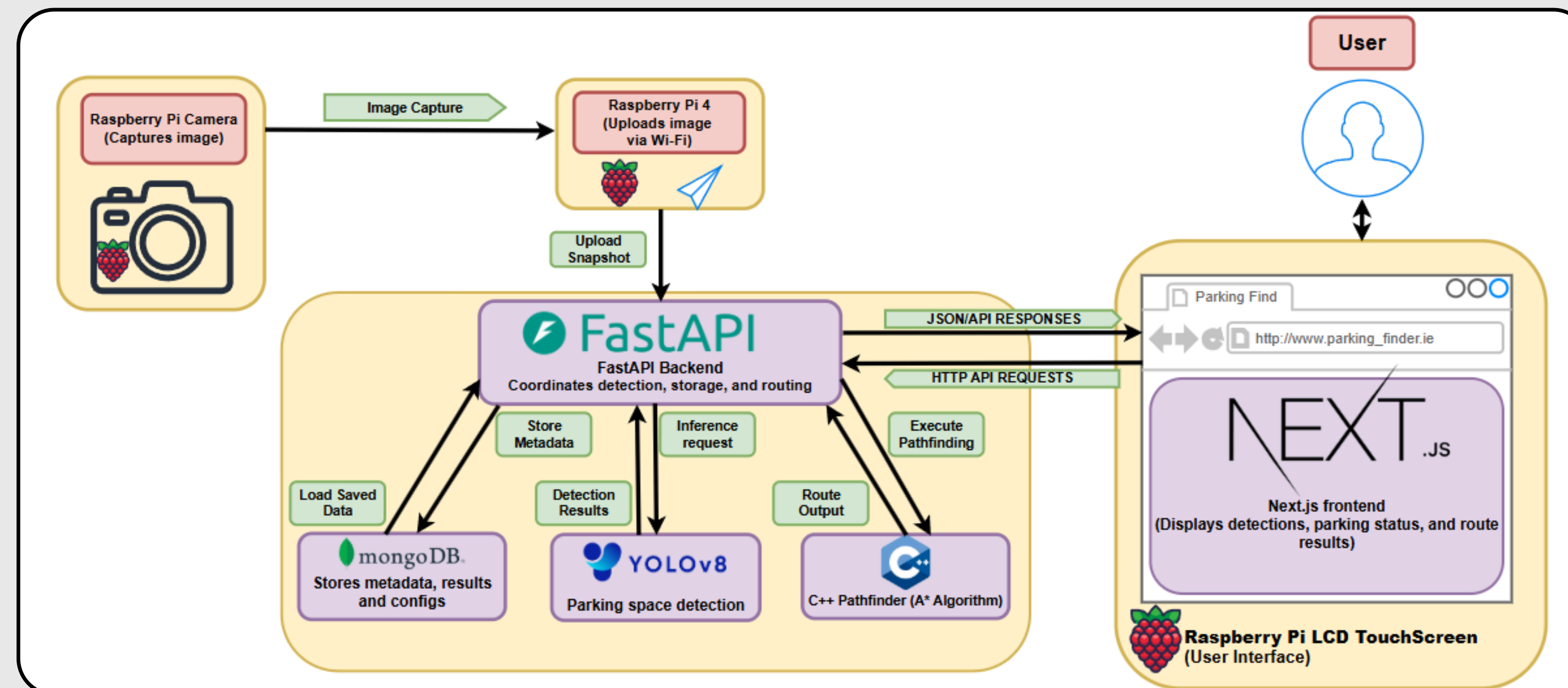
Once free spaces have been identified, a C++ A* pathfinding algorithm calculates the shortest route from the entrance to the nearest available spot. This result is then presented through a web interface using colour coded overlays and route guidance.

PARKING FINDER

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ARCHITECTURE DIAGRAM



HOW IT WORKS

- Raspberry Pi Camera Module 3 captures images at regular intervals.
- The Raspberry Pi 4 uploads the image to the backend over Wi-Fi.
- YOLOv8 detects vehicles in the image.
- Each detected vehicle centre is checked against predefined parking spot polygons.
- Parking spaces are classified as occupied or available.
- Free spaces are mapped onto a 2D grid.
- A C++ A* algorithm calculates the shortest route to the nearest available space.
- The frontend displays the parking status and generated route.

TECHNOLOGIES

Front end: Next.js, React, Type Script, Tailwind CSS

Backend: FastAPI, Python, Uvicorn

AI/ML Vision: YOLOv8, Ultralytics, OpenCV, Pytorch

Database: MongoDB, Motor

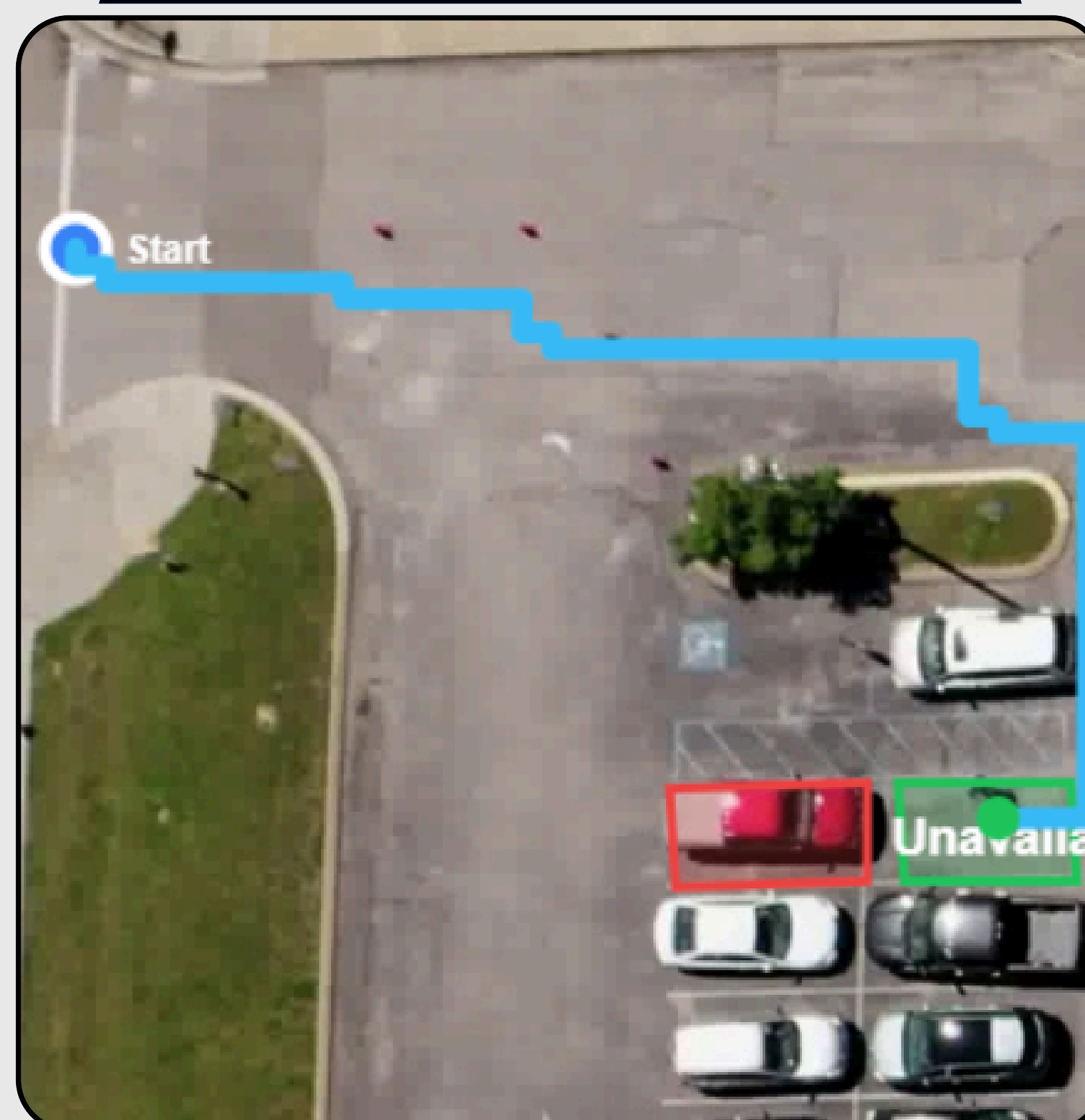
Algorithms: A* Pathfinding, C++

CONCLUSION

This project demonstrates that smart parking can be implemented using AI and software, without relying on costly sensor based infrastructure. It combines vehicle detection, parking space classification, and route generation into one working solution.

Future improvements could include moving from periodic image capture to live video streaming, along with real time campus deployment.

USER VIEW



YOLOv8 RESULTS

